# Lesson 14 Teacher's Key: Raising the Stakes

Godot Engine - Instructor Reference Only

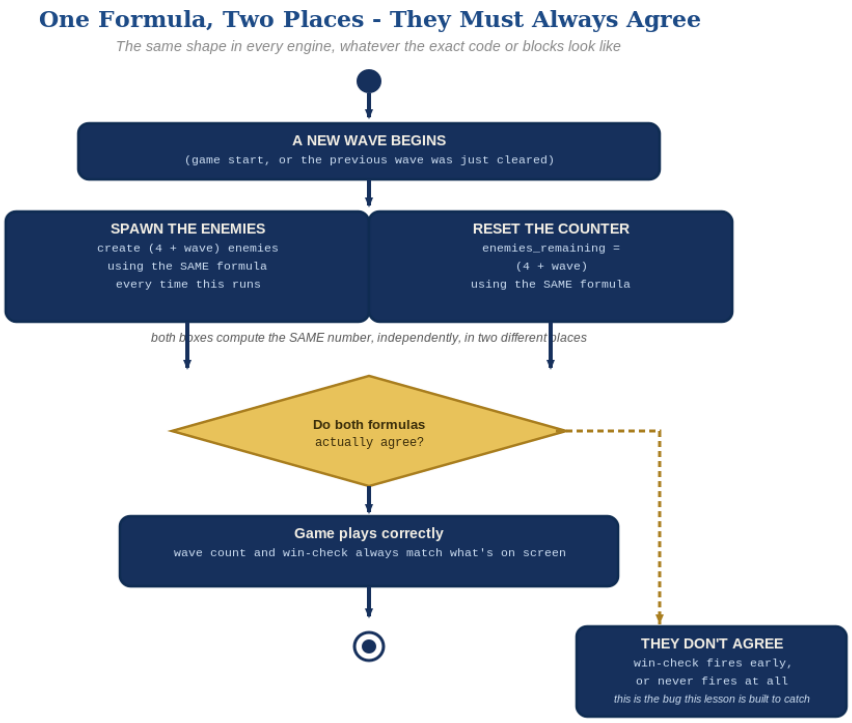
VIDEO GAME CREATION

## What This Lesson Teaches

Same idea as every engine this week - replace hardcoded literals with wave-dependent formulas - expressed as GDScript, split across Main's spawn_enemies() (introduced Lesson 10) and GameManager's own script, exactly where this course has always kept the enemy-creating code and the shared game state, respectively.

## Level 1 - Full Worked Solution

- range(4 + $GameManager.wave) inside spawn_enemies() and enemies_remaining = 4 + wave inside GameManager's two reset spots must all three agree - any mismatch is exactly the kind of bug this Key's troubleshooting section calls out.
- No new variables are needed for Level 1 - this is purely a formula change inside functions that already exist.
- Wave 1 still spawns exactly 5 (4 + 1 = 5), so nothing about the game's opening feel changes - only later waves escalate.




*Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.*

## Why This Structure

- Keeping the loop bound's wave-dependence inside spawn_enemies() (on Main, via $GameManager.wave) while enemies_remaining's reset stays on GameManager itself is exactly Rule 13's existing split - spawn_enemies() is about creating enemies, enemies_remaining is shared game state, and today's change does not blur that line even though both numbers now depend on the same wave.
- For Level 2, setting enemy.speed on each freshly-instantiated enemy (rather than changing enemy.tscn's own default) keeps the scaling logic entirely inside spawn_enemies() - the enemy scene itself stays exactly as simple as Lesson 7 left it.

## Level 2 / Level 3 Talking Points

- Ask students to trace through spawn_enemies() for wave 3 by hand: range(4+3)=range(7) - then separately confirm GameManager's enemies_remaining reset computes the same 7 - a good comprehension check for reading a formula split across two scripts.
- For students comparing engines: point out that Unreal's identical (4 + Wave) formula, expressed as Blueprint math nodes instead of GDScript, is conceptually the exact same idea - a good cross-engine parallel.

## Common Student Mistakes

- Updating spawn_enemies()'s range() but forgetting to update one (or both) of GameManager's enemies_remaining reset spots - or vice versa.
- Forgetting the $GameManager prefix on wave inside spawn_enemies() (it lives on Main, not GameManager) while correctly leaving it off inside GameManager's own start_game()/on_enemy_destroyed() - easy to flip by habit.
- In start_game(), leaving wave = 1 AFTER enemies_remaining's reset instead of before it, so a restart computes enemies_remaining from the previous game's leftover wave number instead of a fresh wave 1.
- Setting enemy.speed BEFORE add_child() instead of after - harmless in Godot 4 either way, but worth confirming the assignment target is actually the enemy just created, not a stale reference.
- Leaving enemy.tscn's default speed := 60.0 as the ONLY place speed is set, forgetting the new per-instance override in spawn_enemies() entirely.